

Worked Answer Key

1. **P1-1:** 3:35; “thirty-five minutes past three.” **P1-2:** 8:45; 15 minutes before 9.
2. **P2-1:** 12:30 a.m. is 30 minutes after midnight and before noon. **P2-2:** 19:45.
3. **P3-1:** 2:35 to 3:35 = 1 h; to 4:10 = 35 min; **1 h 35 min.** **P3-2:** 8:47 to 9:00 = 13 min; plus 32 = **45 min.**
4. **P4-1:** 11:40 + 20 = noon; +35 = **12:35 p.m.** **P4-2:** 10:35 to midnight = 1 h 25 min; +1 h 20 min = **2 h 45 min.**
5. **P5-1:** $3 \times 60 + 18 =$ **198 min.** **P5-2:** $250 = 4 \times 60 + 10$, so **4 h 10 min.**
6. **P6-1:** 9:55 to 10:05 = **10 min.** **P6-2:** 9:10 to 11:25 = **2 h 15 min.**
7. **P7-1:** 18 minutes before 7:10 is 6:52; **6:50 is earlier.** **P7-2:** **12:05 a.m., 11:50 a.m., 12:15 p.m., 11:50 p.m.**
8. **P8-1:** **\$7.06**, or **7.06 dollars.** **P8-2:** $75 + 20 + 20 + 7 =$ **122¢ = \$1.22.**
9. **P9-1:** $12.75 + 8.49 =$ **\$21.24.** **P9-2:** $3.85 + 6.27 =$ **\$10.12.**
10. **P10-1:** $20.00 - 13.68 =$ **\$6.32.** **P10-2:** $35.50 - 18.75 =$ **\$16.75.**
11. **P11-1:** $10.00 - 7.46 =$ **\$2.54.** **P11-2:** $\$20 - \$14.83 =$ \$5.17; e.g., 5-dollar bill, dime, nickel, 2 pennies.
12. **P12-1:** $8.95 + 14.50 + 2.35 =$ **\$25.80.** **P12-2:** $12.49 + 9.75 =$ 22.24; $30 - 22.24 =$ **\$7.76**; yes.
13. **P13-1:** $7.20 \div 4 =$ **\$1.80.** **P13-2:** $5.94 \div 6 =$ **\$0.99.**
14. **P14-1:** $5 \times 6.75 =$ **\$33.75.** **P14-2:** $3 \times 8.40 =$ 25.20; +2.25 = **\$27.45.**
15. **P15-1:** $27.60 \div 4 =$ **\$6.90.** **P15-2:** $45 \div 5 =$ **\$9.00.**
16. **P16-1:** $25 - 8.50 - 6.75 =$ **\$9.75.** **P16-2:** $8 \times 7.25 =$ 58; $100 - 58 =$ **\$42**, enough.
17. **P17-1:** $40 - 8 =$ **\$32.** **P17-2:** tax = $12.50 \times 0.08 =$ **\$1.00**; final = **\$13.50.**
18. **P18-1:** $15 \times 6 =$ **\$90.** **P18-2:** 9:30 + 2 h = 11:30; +45 = **12:15 p.m.**
19. **P19-1:** Change = $25 - 16.80 =$ \$8.20. Given 1 bill + 2 quarters + 3 dimes = 8.80, so **incorrect; it is 60¢ too much.** **P19-2:** 4:15 + 1 h 35 min = **5:50 p.m.**
20. **P20-1:** 9:45 to midnight = 2 h 15 min; +20 min = **2 h 35 min.** Cost $18.75 + 3.60 =$ **\$22.35.** **P20-2:** $2 \times 12.95 =$ 25.90; $3 \times 4.25 =$ 12.75; total 38.65; $50 - 38.65 =$ **\$11.35.**